

through resistor R1. As N1 is configured as an integrator, it outputs a triangular wave of the same frequency as the square wave. The triangular wave is fed to pin 5 of op-amp N3, which is configured as a comparator.

The reference voltage at pin 6 of the comparator is fixed through the potential divider arrangement formed

by potentiometer VR1 and resistors R4 and R5. It can be set from -6V (lowermost position of VR1) to +6V (uppermost position of VR1).

The triangular wave applied at pin 5 of N3 is compared with the reference voltage at its pin 6. The output at pin 7 is about +12V when the voltage at pin 5 is greater than the voltage at pin

6. Similarly, the output at pin 7 is about -12V when the voltage at pin 5 is lower than the voltage at pin 6.

The output from comparator N3 is the gate voltage for n-channel MOSFET (T1). T1 switches on when the gate voltage is positive and switches off when the gate voltage is negative. Setting of the reference voltage therefore controls the pulse-width of the motor.

When T1 is switched on for a longer period, the pulse width will be wider, which means more average DC component and faster speed of the motor. Speed will be low when the pulse width is small. Thus potentiometer VR1 controls the speed of the motor.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. The circuit requires ±12V power supply for its working. It can also be modified to control the speed of a 6V or 24V DC motor. ●

Safe Cigarette Lighter Power Socket

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Cigarette lighter sockets available on the dashboards of some vehicles act as a heat source to light cigarettes. Besides, these can power electronic gadgets like tablets, laptops, portable video players, MP3 players and cellphones. Some of these devices can be plugged directly into the cigarette lighter socket, while others may need an inverter (DC-AC converter).

However, using the cigarette lighter socket to power electronic gadgets while the car engine is not running may drain the car's battery. This circuit lets you use the cigarette lighter socket to power

your electronic gadgets without the fear of draining the car's battery.

The circuit supplies power to electronic devices with low-voltage protection mechanism. A moulded cigarette lighter socket extension cable

(refer Fig. 3) is necessary for the circuit.

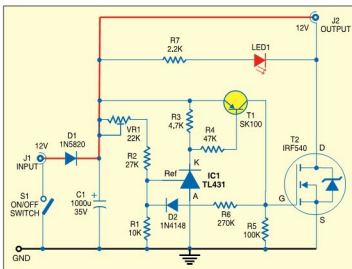


Fig. 1: Safe cigarette lighter socket



Cut the extension cable and solder its leads to the input of the circuit (J1). The circuit receives 12V DC at J1 through polarity-protection diode D1.

As shown in Fig. 1, the heart of the circuit is IC TL431 (IC1)—a voltage regulator wired as a comparator. When the voltage at the reference terminal (Ref) of IC1 exceeds 2.5V, its cathode (K) goes low, which provides base-emitter biasing of transistor T1. Transistor T1 provides regenerative feedback via